

ZENELAIT HRMS

COMPREHENSIVE ROLE-BASED ENTERPRISE MANUAL

An all-inclusive, tier-scoped platform designed for modern workforce management. Built by Zenelait InfoTech Private Limited to streamline workflows from onboarding logistics to real-time payroll analytics.

CORPORATE PROFILE

Zenelait InfoTech is a leading software engineering firm dedicated to building next-generation enterprise applications. We specialize in designing customizable SaaS platforms that integrate HRMS, CRM, and supply chain networks into unified hubs.

ZENELAIT HRMS VISION

Zenelait HRMS enables multi-tenant organizations to achieve 100% data transparency. Our platform bridges the gap between manual attendance logs and corporate auditing, delivering automated compliance checks and high-resolution PDF expense voucher compilations.

OFFICIAL CONTACT & SECURITY NOTICE

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Technical Note: The system utilizes asynchronous Kafka event logs and Redis serialization to ensure zero-latency routing.

SYSTEM ARCHITECTURE

BACKEND REST APIS, FRONTEND ROUTING, AND AUDIT PIPELINES

Zenelait HRMS employs a highly scalable, decoupled client-server architecture model. The client-side dashboard runs on Next.js 14 / Vite, connecting to a multi-threaded Java Spring Boot 3 enterprise backend and a high-performance MySQL/TiDB Cloud relational storage.

TECHNICAL STACK DETAILS

Layer	Technology	Responsibility
Frontend Client	Next.js 14, React, TS, Tailwind CSS	Role Dashboards, Real-time Charts, Responsive Layouts
Java Backend	Spring Boot 3, Java 21, Hibernate JPA	Business Logic, Multi-tenant Data APIs, Security Filters
Messaging Queue	Apache Kafka (Port 9092)	Asynchronous Audit Logging & Event-driven Logs
Cache Broker	Redis Cache (Port 6379)	Performance Optimization & API Response Serialization
Database	MySQL / TiDB Cloud (Port 4000)	Transactional Ledgers, User Accounts, Registry

SECURITY FILTER CHAIN & DECOUPLING

All network interactions use JSON-encoded requests. Spring Security handles endpoint protection, verifying signature validation keys encoded in client-header HTTP bearer authentication tokens on every dispatch.

The backend employs database connection pools (HikariCP) to interact with relational schemas, ensuring fast transactional speeds during high-concurrency workforce attendance logging.

SUBSCRIPTION TIERS

MODULAR CONTROL AND PRICING PLAN MAPPINGS

To provide flexible enterprise SaaS packages, Zenelait HRMS employs an active subscription mapping engine. The features, dashboards, and role dropdowns rendered on login screens are toggled dynamically based on the active client subscription tier.

GROWTH PLAN

Designed for small operations. Unlocks employee clock-in/out registers and self-service. Excludes deep financial ledger control.

PREMIUM PLAN

Adds middle-tier control. Unlocks full HR Manager control panels, Sprint velocity trackers, custom timesheets, and basic payroll metrics.

ENTERPRISE PLAN

Full platform access. Enables the Super Admin console, multi-tenant custom organization settings, Kafka audit streams, and Redis cache tuning.

DATABASE CONFIGURATION & VERIFICATION

The `organizations` table stores subscription levels under the `plan\_type` column (STANDARD, MIDDLELEVEL, ENTERPRISE). When a user authenticates, the handler queries `GET /api/superadmin/organizations` to fetch their tenant context, parsing the subscription plan type to dynamically configure permission matrices on the frontend client.

ACCESS SECURITY

OTP LIFECYCLES, PORTAL SELECTION, AND AUTH FLOWS

1. THE FIREWALL PORTAL SELECTOR GATE

Access control starts at the landing page. Before hitting the authentication pages, the portal selection panel remains locked behind an organization name, code, and OTP verification modal. This acts as a primary firewall.

2. OTP LIFECYCLE

- **Step A:** The user inputs their corresponding organization name, which queries ``GET /api/superadmin/organizations``.
- **Step B:** The system verifies the organization code and requires the 6-digit OTP code corresponding to the tenant.
- **Step C:** On verification, organization configurations (ID, Plan Tier, Code) are committed to client `localStorage`, unlocking role portals.

3. LOGIN GATEWAYS

Once unlocked, users are directed to role-specific login interfaces: ``/login/superadmin``, ``/login/admin``, or ``/login/erp``. All forms submit username and password details via ``POST /api/auth/login``, returning authenticated user models.

SUPER ADMIN CONSOLE

TENANT PROVISIONING, SUBSCRIPTION BILLING & HEALTH STATUS

1. TENANT PROVISIONING

The Super Admin is the primary systems manager. Using the provision form, they register new organizations into MySQL. The system automatically generates a unique Organization Code (e.g. HRMS2026xxxxx) and a 6-digit OTP code.

2. ACTIONS INVOLVED

- **GET /api/superadmin/organizations** Fetches the complete registry of active tenant companies.
- **POST /api/superadmin/organizations** Submits organization configuration payload (name, type, owner contact, billing plan).
- **PUT /api/superadmin/organizations/{id}** Updates active tier specifications or details.
- **DELETE /api/superadmin/organizations/{id}** Removes a tenant from the registry.

3. KAFKA EVENT QUEUES

Every organization creation triggers a Kafka message to `organization-creation-events` detailing ID, name, code a

EMPLOYEE ROSTER MANAGEMENT

HR ADMIN CONTROL OF DIRECTORIES AND ACCOUNT CREATIONS

1. THE HR ROSTER INTERFACE

HR Administrators (Tenant Admins) manage the active roster for their organization. The screen presents a creation form and an active roster grid containing user profile coordinates.

2. ACTIONS INVOLVED

- **GET /api/auth/users?orgId={id}** Fetches employee records registered under the Admin's organization.
- **POST /api/auth/create-user** Creates a new employee account in MySQL, setting username, password, email, and mobile.
- **PUT /api/auth/users/{id}/role** Modifies an employee's role level (e.g. Admin, Employee).

3. SECURITY NOTICE

New registrations default to role `EMPLOYEE` or `ADMIN`, automatically bound to the parent organization ID.

RECRUITMENT: JOB REQUISITIONS

CREATING OPENINGS, ASSIGNING TARGET DEPARTMENTS

1. THE JOB REGISTER SCREEN

The recruitment module starts with job requisitions. HR Managers can create and track positions across departments like Engineering, Sales, and Human Resources. Status toggles indicate open or closed pipelines.

2. ACTIONS INVOLVED

- **GET /api/recruitment/jobs?orgId={id}** Fetches all job requisitions for the organization.
- **POST /api/recruitment/jobs** Submits a new job opening specifying title, target department, and initial status.

3. DATABASE PERSISTENCE

Job requisitions are stored in the `job\_requisitions` table, establishing relations with candidate records.

RECRUITMENT: CANDIDATES PIPELINE

MANAGING APPLICATIONS AND PIPELINE PIPELINES

1. THE INTERACTIVE PIPELINE BOARD

The Candidate Pipeline presents a visual board layout where applicants are categorized into columns representing stages: Applied, Screening, Interview, Offered, Hired, and Rejected. Drag-and-drop operations trigger updates.

2. ACTIONS INVOLVED

- **GET /api/recruitment/candidates?orgId={id}** Reads all candidate profiles registered under the organization.
- **POST /api/recruitment/candidates** Adds a new candidate record mapped to a specific job requisition.
- **POST /api/recruitment/candidates/stage** Updates the candidate's stage status in MySQL when moved in the board layout.

3. PRE-SEEDED DATA MAPPING

Profiles like Alice Smith (Applied), Bob Johnson (Screening), and Charlie Brown (Interview) are persisted in `candi

ATTENDANCE & TIMEKEEPING

CLOCK-IN/OUT LIFECYCLE AND LOG AUDITING

1. CLOCK TRIGGERS

Employees log work time on their dashboard using the 'Clock In' and 'Clock Out' button. The action registers exact times, calculating daily duration and compliance metrics automatically.

2. ACTIONS INVOLVED

- **GET /api/attendance?orgId={id}** Fetches the full company log history (Admin view).
- **POST /api/attendance/clock** Submits clock-in or clock-out timestamp events for the current user.
- **POST /api/attendance/save?orgId={id}** Bulk saves manual grid entry adjustments made by HR.

3. METRICS AND CHARTS

Daily active employee stats and monthly attendance rates are graphed on the leadership dashboards.

LEAVE MANAGEMENT

SUBMITTING CLAIMS, BALANCE TRACKING, AND APPROVAL TREE

1. THE LEAVE WORKFLOW

Employees can submit requests for Casual, Sick, or WFH leaves. HR Admins view all pending requests in an approval tree, allowing them to accept or reject them, which updates the employee's remaining leave balance.

2. ACTIONS INVOLVED

- **GET /api/leaves?username={name}**: Loads historical leave requests and balances for the logged-in user.
- **POST /api/leaves**: Creates a new leave request (date, duration, type).
- **POST /api/leaves/approve**: Submits approval or rejection state update for a pending ID.

3. DATABASE SCHEMA

Leaves are persisted in `leave\_requests` table. Remaining balances are calculated dynamically from approved du

PROJECT PORTFOLIO

PROJECT REGISTRY AND PORTFOLIO CONTROL

1. PORTFOLIO MONITORING

Zenelait HRMS tracks project portfolios. Managers register projects, allocate budgets, assign team owners, and evaluate status flags (Green, Amber, Red). Visual Gantt schedules display milestones.

2. ACTIONS INVOLVED

- **GET /api/projects?orgId={id}** Loads the registry of projects mapped to the organization.
- **POST /api/projects** Creates a new project record, storing name, budget limit, and initial status.
- **PUT /api/projects/{id}** Updates project parameters or tracks financial spend.

3. DATA ENTITY

Data mapping connects the `projects` table directly to financial gauges displayed on the management dashboards

SPRINTS & SCRUM BACKLOG

SPRINT LIFECYCLES, SCRUM TICKETS AND VELOCITY BOARDS

1. SPRINT LIFECYCLE

Teams organize engineering tasks into Sprints. The Backlog interface displays scrum tickets categorized by status: To Do, In Progress, In Review, and Done. Team members can move tickets to track daily velocity.

2. ACTIONS INVOLVED

- **GET /api/sprints?orgId={id}** fetches sprint milestones and active timelines.
- **POST /api/sprints/create?orgId={id}** creates a new sprint cycle.
- **GET /api/tickets?orgId={id}** loads all scrum task tickets for the active sprint.
- **POST /api/tickets/create?orgId={id}** submits a new ticket, setting title, description, points, and assignee.

3. SPRINT BOARDS

Dragging tickets automatically updates active project completion indicators in the database.

PAYROLL CALCULATIONS

PROCESSING PAYROLL LOGS AND GENERATING STATEMENTS

1. PAYROLL PROCESSING

HR Admins execute payroll calculations at the end of each payment cycle. The system evaluates active employee logs, calculates gross salary, applies tax deductions, and generates printable payroll registers.

2. ACTIONS INVOLVED

- **GET /api/payroll?orgId={id}** loads historical payroll statements.
- **GET /api/payroll/calculate?orgId={id}** simulates active wage runs, applying tax rules and deduction formulas.
- **POST /api/payroll/process?orgId={id}** locks and records the payroll run into the `payroll` database table.

3. KAFKA EVENT PIPELINE

Successful payroll runs trigger asynchronous transaction logs dispatched to the system auditing pipeline.

CORPORATE EXPENSES & VOUCHERS

SUBMITTING REIMBURSEMENTS AND DYNAMIC PDF GENERATION

1. THE VOUCHER GENERATOR

Employees submit claims for business expenses (travel, meals, assets). The Expense ledger contains an overlay panel to preview, compile, and download high-resolution, pixel-perfect PDF vouchers with signature blocks.

2. ACTIONS INVOLVED

- **GET /api/expenses?orgId={id}**: Loads all submitted expense reimbursement claims.
- **POST /api/expenses**: Submits a new claim, attaching receipt URL links and category descriptions.
- **PUT /api/expenses/{id}/status**: Approves or rejects claims (HR Admin control).

3. PDF COMPILER TECH

The client utilizes html2canvas and jspdf to capture the DOM template at scale: 2 for clean, print-ready document

INVOICES & RECEIVABLES

CLIENT BILLINGS, AUTO-GENERATION AND AGING MATRIX

1. BILLING AND INVOICING

The Invoices ledger tracks accounts receivables from corporate clients. HR Admins can manually register invoices or trigger auto-generation schedules based on contract targets. Status flags trace payment delays.

2. ACTIONS INVOLVED

- **GET /api/invoices?orgId={id}** loads the registry of client invoices from the database.
- **POST /api/invoices/create?orgId={id}** creates a manual invoice record in MySQL.
- **POST /api/invoices/auto-generate?orgId={id}** runs automated contract target evaluations.

3. RECEIVABLES ENGINE

Payment records update cash position charts mapped on the Finance dashboard.

PERFORMANCE APPRAISALS

CREATING REVIEW CARDS AND AUDITING RATING MATRICES

1. PERFORMANCE MANAGEMENT

Managers evaluate employee performance. Review dashboards display appraisal cards mapping key metrics: Technical proficiency, communication skills, delivery rates, and overall manager recommendations.

2. ACTIONS INVOLVED

- **GET /api/performance?orgId={id}** Fetches review cards logged for organization employees.
- **POST /api/performance** Creates a review record, storing employee ID, rating points, and text feedback.

3. PERFORMANCE AUDITING

Average ratings are calculated in the backend, updating active performance indexes on the dashboard.

ASSETS & HELPDESK

ASSET ALLOCATIONS AND SUPPORT TICKETS PORTAL

1. CORPORATE ASSETS INVENTORY

Employees can request corporate hardware (laptops, monitors). HR tracks hardware inventory statuses: Allocated, Available, Pending.

2. ACTIONS INVOLVED

- **GET /api/assets?orgId={id}** Fetches hardware asset list from the database.
- **POST /api/assets/request** Submits a device request from an employee profile.
- **GET /api/helpdesk?orgId={id}** Loads support tickets logged by employees.
- **POST /api/helpdesk/create** Logs a technical issue (setting description, category, and priority).

3. SUPPORT QUEUES

Helpdesk tickets are routed to support queue lines in the `helpdesk\_tickets` table.

EMPLOYEE SELF-SERVICE

DASHBOARD CONTROLS, WORK LOGS AND PASSWORD RESETS

1. THE SELF-SERVICE SCREEN

Employees log daily work details via the Self-Service interface. From their personal command console, they register attendance, review salary history, track task logs, and submit password reset requests.

2. ACTIONS INVOLVED

- **GET /api/worklogs?username={name}** Loads daily task details logged by the user.
- **POST /api/worklogs** Submits a daily log entry (hours, task description).
- **POST /api/auth/forgot-password** Dispatches password reset request to HR queue.

3. SYSTEM SECURITY

Users change credentials using `POST /api/auth/change-password``, requiring current password verification.

DEVELOPER REGISTRY

CORE BACKEND REST APIS AND DATABASE CONSTRAINTS

1. DATABASE ENTITIES REFERENCE

The schema database contains 31 JPA entity tables with foreign keys mapping multi-tenant organization IDs. Auto-generating columns match Java serialVersionUID specifications for secure database transactions.

2. CORE API ENDPOINTS MAPPING

- **GET /api/superadmin/organizations** System-wide tenant registries query.
- **POST /api/auth/login** User login credentials verification endpoint.
- **GET /api/recruitment/candidates**: Recruitment pipeline details query.
- **GET /api/attendance**: Daily work log attendance queries.
- **GET /api/leaves**: Leaves balance and application registers.

3. RELATIONAL INTEGRITY Constraints

User mappings enforce cascade exclusions to prevent tenant data leakage. Every transaction updates MySQL.